A LiDAR Odometry With Visual Semantic Segmentation-assisted Dynamic Object Removal

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***Abstract*—Accurate and robust vehicle self-localization is crucial for autonomous driving. However, the static world assumption used in traditional lidar odometry methods is violated due to dynamic objects (such as pedestrians and vehicles) common in urban environments, leading to degraded localization accuracy. Although dynamic objects have been tracked and removed in recent approaches using LiDAR-based detection front-ends, missed detections still frequently occur in scenarios involving large-scale close-range occlusions or sparse point clouds at long distances. To address these challenges, this paper proposes a LiDAR odometry framework with visual semantic segmentation-assisted dynamic object removal. Firstly, a LiDAR-based object detection front-end is employed to preliminarily eliminate dynamic LiDAR points within detected 3D bounding boxes. Then, a two-stage dynamic point removal strategy is conducted by leveraging parallel visual semantic segmentation results. In the first stage, LiDAR points are projected onto the camera image plane and those falling within dynamic object masks are identified as candidate dynamic points. In the second stage, candidate points located inside static 3D detection bounding boxes are preserved to avoid discarding valid measurements, while the remaining points are removed as dynamic outliers. The resulting static point cloud, together with 3D detection bounding boxes, is finally fed into the LiDAR odometry module to estimate accurate ego-vehicle poses. Experimental results demonstrate the effectiveness and superiority of the proposed method in dynamic urban environments.**

***Index Terms*—LiDAR odometry, dynamic object removal, visual semantic segmentation, vehicle self-localization.**

I. Introduction

Accurate and robust self-localization is a fundamental requirement for autonomous driving systems [1]–[5]. Existing localization approaches can be broadly categorized into vision-dominant odometry [6], [7] and light detection and ranging (LiDAR)-dominant odometry [8]–[10]. Vision-based methods benefit from rich semantic information but are often sensitive to illumination changes, motion blur, and texture sparsity [6]. In contrast, LiDAR-based odometry directly measures precise geometric structures of the environment and is largely invariant to lighting conditions. As a result, LiDAR-centric methods generally achieve higher accuracy, stronger robustness, and better adaptability across diverse driving scenarios, including urban roads, highways, and complex outdoor environments [2]. Owing to these advantages, LiDAR-based odometry (LO) has become a more favorable solution for reliable vehicle self-localization in real-world autonomous driving applications.

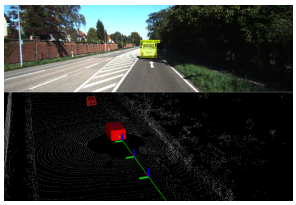
Although existing LiDAR odometry (LO) methods have demonstrated strong performance in static and structured environments, their accuracy degrades significantly in dynamic urban scenarios [11]. This is mainly because the presence of moving objects violates the static-world assumption underlying most traditional LO pipelines. To address this issue, several recent approaches incorporate LiDAR-based 3D dynamic object tracking and removal to mitigate the influence of moving objects [12]–[14]. However, representative 3D object detection networks, such as CenterPoint [15], may suffer from missed detections when facing large dynamic objects under close-range occlusion or distant observations, as shown in Fig. 1. As a result, dynamic LiDAR points may be mistakenly treated as static measurements, making it difficult to completely eliminate dynamic interference and thereby limiting localization accuracy.

Compared with LiDAR point cloud–based 3D object detection, visual information provides much richer semantic cues [5]. As a result, vision-based methods are capable of accurately segmenting dynamic objects under challenging conditions, such as large-scale close-range occlusions and distant observation, as illustrated in Fig. 1. Motivated by this observation, this paper proposes a LiDAR odometry framework with visual semantic segmentation–assisted dynamic object removal. Specifically, a LiDAR-based 3D object detection front-end is first utilized to coarsely filter out dynamic LiDAR

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(a) Large-scale dynamic object occlusions



(b) Long-range dynamic objects

Fig. 1. Demonstration of dynamic LiDAR point removal via visual semantic segmentation. The blue bounding boxes indicate LiDAR-based 3D object detection results, while the red points denote dynamic LiDAR points that are not correctly identified by the LiDAR detection front-end due to large-scale occlusions by dynamic objects or long-range observations. In contrast, visual semantic segmentation can accurately identify these dynamic LiDAR points and effectively remove them.

points enclosed within detected dynamic bounding boxes. Subsequently, visual semantic segmentation results, obtained in parallel, are leveraged to further refine dynamic point removal through a two-stage strategy. In the first stage, LiDAR points are projected onto the image plane and those associated with dynamic semantic masks are marked as potential dynamic candidates. In the second stage, these candidate points are examined against static 3D detection bounding boxes: points falling inside static object boxes are retained to prevent the loss of valid measurements, while the remaining points are regarded as dynamic outliers and removed. Finally, the refined static point cloud, together with the 3D detection bounding boxes, is fed into the LiDAR odometry back-end to produce accurate ego-vehicle pose estimates. The proposed method is developed upon the TRLO framework [14], and its main contributions are summarized as follows:

1) A LiDAR odometry method with visual semantic segmentation–assisted dynamic LiDAR point removal is proposed. The method is capable of tracking and eliminating dynamic objects, and further refines dynamic LiDAR point removal by incorporating a lightweight visual semantic segmentation network, thereby enabling accurate and robust vehicle self-localization in dynamic urban driving environments.
2) A two-stage visual semantic segmentation–based dynamic LiDAR point removal strategy is designed, which can accurately eliminate dynamic points missed by the LiDAR-based 3D object detection front-end, while simultaneously avoiding the removal of valid static points.
3) The effectiveness and superiority of the proposed method are validated through comparisons with several existing state-of-the-art methods on public datasets.

II. Related Works

A. Traditional LiDAR Odometry

LiDAR odometry (LO) has been extensively studied and serves as a fundamental solution for accurate vehicle localization [8]–[10]. Early approaches are mainly feature-based and optimization-driven, with LOAM being a representative framework that extracts edge and planar features to perform scan-to-scan odometry and scan-to-map refinement in a multi-threaded manner [16]. Many subsequent LiDAR-only variants follow this paradigm, focusing on improving robustness and computational efficiency [17], or primarily emphasizing scalable map representations, which leverages structures such as probabilistic voxel maps [10] to enable efficient localization in large-scale environments while maintaining competitive accuracy.

Overall, traditional LO methods have demonstrated excellent performance in static and structured environments. However, their underlying static-world assumption limits robustness in dynamic urban scenarios, motivating further research toward dynamic-aware localization frameworks.

B. LiDAR-based Odometry in Dynamic Environments

Traditional LO methods generally rely on the static-world assumption and thus suffer from degraded performance in dynamic environments, where moving objects such as vehicles and pedestrians introduce inconsistent measurements. With the rapid advancement of deep learning in scene understanding, several approaches have attempted to explicitly identify and remove dynamic LiDAR points to improve robustness [11], [12], [14]. Early efforts leverage LiDAR-based semantic segmentation to classify each point and filter out dynamic categories. For instance, SuMa++ [11] employs RangeNet++ [18] to perform spherical projection–based semantic segmentation and removes potentially dynamic points by inferring per-point semantic probabilities. However, such LiDAR semantic segmentation methods typically require high-resolution and densely populated LiDAR sensors, resulting in increased hardware cost and limited practicality.

More recent approaches focus on LiDAR-based 3D object detection and tracking to mitigate dynamic interference. LIMOT [12] builds upon LIO-SAM [9] and integrates CenterPoint [15] as a 3D object detection front-end to track dynamic objects, while jointly optimizing ego-motion and object states within a factor graph framework. Similarly, TRLO [14] adopts CenterPoint [15] to detect dynamic objects and removes LiDAR points belonging to these objects, enabling accurate self-localization and consistent mapping in urban environments. Despite their effectiveness, these detection-driven methods may suffer from missed detections when large dynamic objects cause close-range occlusions or when targets are observed at long distances with sparse point clouds. Consequently, undetected dynamic points may still be treated as static measurements, leading to residual dynamic interference and degraded localization accuracy.

III. Methods

A. Notation and System Overview

Three coordinate frames are considered in this paper. The LiDAR/body frame associated with the k-th LiDAR scan at time t_k is denoted by L_k, the camera frame at time t_k is denoted by C_k, and the world frame is denoted by W. Following a similar definition in [12], let $\mathbf{T}_b^a \in \mathrm{SE}(3)$ denote the pose of

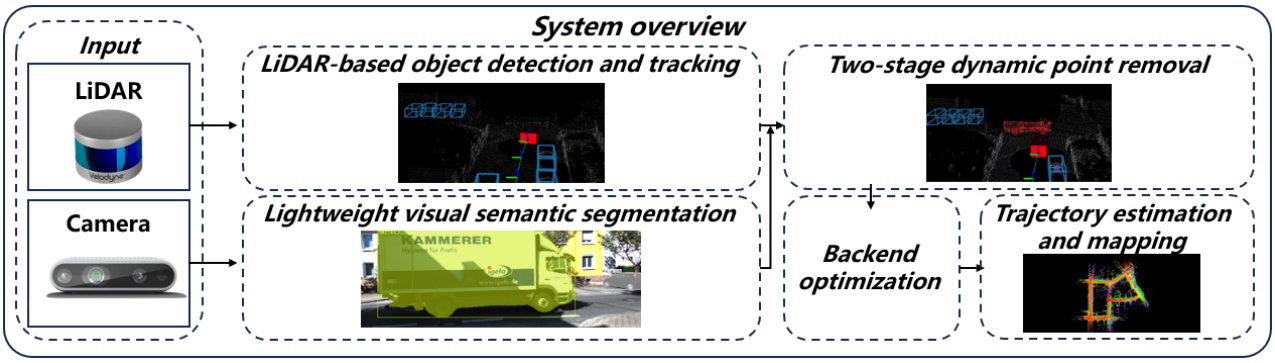


Fig. 2. Overview of the proposed method. The blue bounding boxes represent the 3D object detection results based on LiDAR point clouds, the red box represents the ego vehicle, and the red point cloud represents dynamic LiDAR points that are not identified by the object detection front-end but are detected and removed by the visual semantic segmentation.

frame b expressed in frame a. Consequently, $\mathbf{T}_{L_k}^{W}$ represents the pose of the ego-vehicle in the world frame at time t_k, while $\mathbf{T}_{L_{k-1}}^{L_k}$ represents the relative pose transformation of the ego-vehicle from t_{k-1} to t_k. For convenience, $\mathbf{T}_{L_k}^{W}$ and $\mathbf{T}_{L_{k-1}}^{L_k}$ are hereafter denoted as $\mathbf{T}_k^W$ and $\mathbf{T}_{k-1}^k$, respectively. In addition, $\mathbf{T}_k^W$ consists of a translation component $\mathbf{t}_k \in \mathbb{R}^3$ and a rotation component $\mathbf{R}_k \in \mathrm{SO}(3)$.

Consider a vehicle driving on urban roads, equipped with a camera and LiDAR. Since moving vehicles, pedestrians, and other dynamic objects violate the static-world assumption of traditional LiDAR SLAM, they can severely degrade its accuracy and robustness. Therefore, the proposed method mainly aims to address the following problem: given the LiDAR point cloud $\mathcal{P}^k$ in L_k frame, dynamic-object points are detected and removed to obtain a static LiDAR point cloud $\mathcal{P}_{static}^k$ as input to the LO, ultimately yielding an accurate pose estimate of the ego-vehicle. The pipeline of the proposed method is illustrated in Fig. 2. Existing approaches typically perform dynamic feature removal using LiDAR-based object detection and tracking. However, large-area occlusions in near-range LiDAR point clouds or sparsity in far-range point clouds may lead to missed detections, making it difficult to completely remove dynamic points. Unlike existing methods, the proposed approach runs a lightweight visual semantic segmentation network in parallel to accurately detect objects that are missed due to severe occlusions or long-range sparsity, thereby enabling complete removal of dynamic point clouds.

B. LiDAR-based 3D Object Detection and Tracking

The implementation of LiDAR-based 3D object detection and tracking follows a pipeline similar to that of TRLO [14]. Specifically, PointPillars is adopted as the backbone for LiDAR point cloud feature extraction, and CenterPoint is integrated in the front end to generate 3D detection bounding boxes for moving vehicles or pedestrians. In addition, $\mathbf{T}_{k,O_j}^{W}$ denotes the pose of the j-th 3D detection bounding box in the LiDAR frame, and it is noted that $\mathbf{T}_{k,O}^{W} = \{\mathbf{T}_{k,O_j}^{W} | j = 1, 2..., J\}$, where J dynamic objects are detected at time t_k. In addition, the proposed method employs an Unscented Kalman Filter (UKF) to perform nonlinear prediction and tracking of the motion states of dynamic objects, and adopts a nearest-neighbor (NN) strategy for data association. In addition, dynamic and static objects are distinguished using a velocity threshold $v_{threshold}$, and further details can be found in [14]. The proposed method employs a 360° LiDAR, which provides a wider field of view compared to a camera. Therefore, before removing dynamic points based on visual semantic segmentation results, an initial removal of points within the 3D detection boxes of dynamic objects is performed based on the results of object detection and tracking.

C. Visual Segmentation-based Dynamic Point Removal

CenterPoint adopts a bird's-eye-view (BEV)–based approach, in which point clouds are projected along the z-axis for feature extraction. However, this strategy may fail to extract effective features in the $x-o-y$ plane under large-area occlusions caused by nearby objects, leading to missed detections, as illustrated in Fig. 1. In addition, sparsity of point clouds at long ranges may also result in missed detections. Compared with LiDAR-based object detection, image-based semantic segmentation provides denser and more discriminative object-level cues and can still accurately segment potential dynamic objects under severe occlusions. Therefore, visual semantic segmentation is leveraged to further remove the remaining dynamic points that are missed by LiDAR-based detection. To ensure real-time performance and computational efficiency, the proposed method adopts a lightweight YOLOv8-seg [19], which is executed in parallel to provide accurate and efficient visual semantic segmentation results. Assuming that the image width and height are W and H, respectively, the semantic segmentation mask is obtained as $\boldsymbol{M}^{C_k} \in \mathbb{Z}^{H \times W}$.

The proposed method adopts a two-stage verification scheme to determine whether LiDAR points are dynamic. Firstly, LiDAR points are projected onto the camera image plane, and it is checked whether they fall within the semantic segmentation masks of potential dynamic objects (e.g., vehicles and pedestrians). Consider the i-th LiDAR point at time t_k, whose homogeneous coordinates are given by $\boldsymbol{\mathcal{P}}_i^{L_k} = [x_i^{L_k},\ y_i^{L_k},\ z_i^{L_k},\ 1]^\top$. The point $\boldsymbol{\mathcal{P}}_i$ is transformed

from the L_k frame to the C_k frame as

$$\boldsymbol{\mathcal{P}}_i^{C_k} = \mathbf{T}_{L_k}^{C_k} \boldsymbol{\mathcal{P}}_i^{L_k} = [x_i^{C_k},\ y_i^{C_k},\ z_i^{C_k},\ 1]^\top \tag{1}$$

where $\mathbf{T}_{L_k}^{C_k}$ denotes the relative pose transformation from LiDAR frame to camera frame. Since the LiDAR and the camera are rigidly mounted, the transformation $\mathbf{T}_{L_k}^{C_k}$ can be obtained in advance through calibration. Subsequently, the point expressed in the camera coordinate frame is projected onto the image plane using the camera intrinsic matrix $\mathbf{K} \in \mathbb{R}^{3\times3}$. The intrinsic matrix can be obtained through calibration and is expressed as

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

where f_x and f_y denote the focal lengths, and (c_x, c_y) denotes the principal point. The projection is computed as

$$\begin{bmatrix} u_i' \\ v_i' \\ w \end{bmatrix} = \mathbf{K} \begin{bmatrix} x_i^{C_k} \\ y_i^{C_k} \\ z_i^{C_k} \end{bmatrix} \tag{3}$$

where u_i' and v_i' denote the homogeneous coordinates of the point projected onto the image plane along the u-axes and v-axes, respectively, and w denotes the depth. Finally, the corresponding normalized pixel coordinates can be obtained as

$$u_i = \frac{u_i'}{w} = \frac{f_x \cdot x_C}{z_C} + c_x, v_i = \frac{v_i'}{w} = \frac{f_y \cdot y_C}{z_C} + c_y \tag{4}$$

Afterward, it is checked whether the normalized pixel coordinates (u_i, v_i) fall within the semantic segmentation mask $\boldsymbol{M}^{C_k}$. If there is $\boldsymbol{M}^{C_k}(u_i, v_i) \in \mathcal{C}_{dynamic}$, the i-th LiDAR point $\boldsymbol{\mathcal{P}}_i$ is classified as a candidate dynamic point.

If points projected into the masks of potential dynamic objects are directly removed, points belonging to static objects may also be mistakenly discarded, resulting in the loss of valid measurements. Therefore, in the second stage, candidate dynamic points are further evaluated by incorporating the results of LiDAR-based 3D object detection and tracking. Specifically, if a candidate dynamic point lies within a 3D detection bounding box that is classified as static, it is regarded as a static point. Otherwise, it is considered to belong to a missed dynamic object and is subsequently removed. Finally, a set of static LiDAR point cloud $\mathcal{P}_{static}^{L_k}$ is obtained and used as the input to the LO.

D. LiDAR Odometry

The LO aims to obtain an optimal estimate of the current ego-vehicle pose $\mathbf{T}_k^W$ by exploiting the static LiDAR point cloud $\mathcal{P}_{static}^{L_k}$ together with the pose of 3D object detection bounding boxes $\mathbf{T}_{k,O_j}^W$. This can be achieved by constructing and minimizing three types of residual terms, including LiDAR scan registration residuals, 3D detection bounding box consistency residuals, and local z-axis residuals.

The LiDAR scan registration residual is constructed as

$$res_{scan}(\mathbf{T}_k^W, \mathcal{P}_{static}^{L_k}, \mathcal{S}_k^W) = \sum_i^I \tilde{\mathcal{P}}_i^\top [\Sigma_{k,i}^S + (\mathbf{T}_k^W)^\top \Sigma_{k,i}^S \mathbf{T}_k^W] \tilde{\mathcal{P}}_i \tag{5}$$

where $\mathcal{S}^W$ denotes the local submap at time t_k, which can be referred to [14]. I denotes the number of corresponding points between $\mathcal{P}_{static}^{L_k}$ and $\mathcal{S}_k^W$, $\Sigma_{k,i}^S$ denotes the covariance matrix of i-th associated point in the local submap. In addition, there is $\tilde{\mathcal{P}}_i = \mathcal{P}_i^S - \mathbf{T}_k^W \mathcal{P}_i^{L_k}$, where $\mathcal{P}_i^S$ denotes the homogeneous coordinate of the i-th point in the local submap.

Following the same implementation in [14], the 3D detection bounding box consistency residuals and local z-axis residuals are incorporated to optimize the roll and pitch of the ego-vehicle, as well as the drift along the z-axis. The detected objects are assumed to lie on the same flat plane, with the center point at the bottom of the 3D detection bounding box used to fit the ground normal vector model n^g. The ego-vehicle is assumed to lie on the same plane, allowing n^g to be used to constrain the roll and pitch of the ego-vehicle. The 3D detection bounding box consistency residual is constructed as

$$res_{obj}(\mathbf{T}_k^W, \mathbf{T}_{k,O}^W) = \Gamma(log(\mathbf{R}_k^W)_{r,p}, n^g) \tag{6}$$

where $\Gamma(\cdot)$ denotes the angle difference function and $(\cdot)_{r,p}$ indicates that only the roll and pitch components are considered. To constrain the drift along the z-axis, if the average variation of the detected objects along the z-axis between two consecutive frames is smaller than $0.1\,\mathrm{m}$, the ego-vehicle is assumed to have no displacement along the z-axis. Accordingly, the following constraint is constructed as

$$res_z(\mathbf{T}_k^W, \mathbf{T}_{k-1}^W, \mathbf{T}_{k,O}^W) = d(t_{k,z} - t_{k-1,z}) \tag{7}$$

where $d(\cdot)$ denotes the distance measurement function, $t_{k,z}$ denotes the z-axis coordinate of the ego-vehicle at time t_k. Above all, the optimization problem can be denoted as

$$\begin{aligned} \hat{\mathbf{T}}_k^W = \underset{\mathbf{T}_k^W}{arg\ min} \{ & res_{scan}(\mathbf{T}_k^W, \mathcal{P}_{static}^{L_k}, \mathcal{S}_k^W) \\ & + res_{obj}(\mathbf{T}_k^W, \mathbf{T}_{k,O}^W) + res_z(\mathbf{T}_k^W, \mathbf{T}_{k-1}^W, \mathbf{T}_{k,O}^W) \} \end{aligned} \tag{8}$$

where $\hat{\mathbf{T}}_k^W$ denotes the estimation of $\mathbf{T}_k^W$.

IV. Experiments

In this section, the effectiveness and superiority of the proposed method are validated using real-world datasets. Several representative sequences from the KITTI dataset [20] are selected. The experimental results clearly demonstrate that the proposed method achieves accurate and robust self-localization in dynamic urban road environments.

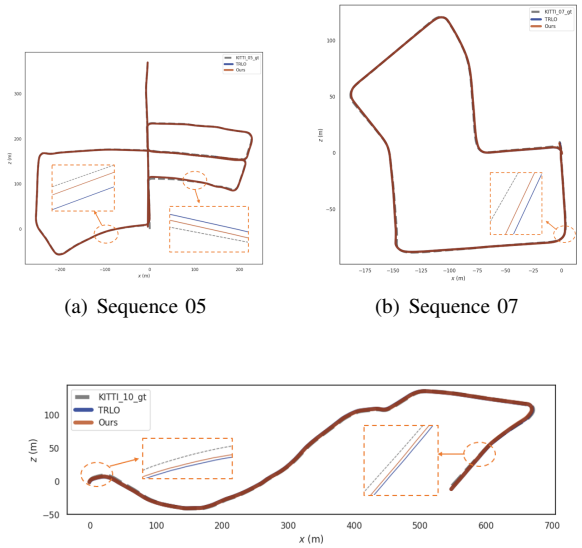
A. Testing Illustrations and Evaluation Criteria

1) Testing Illustrations: The proposed method is compared with the conventional LiDAR odometry method DLO [21], and TRLO [14], which performs LiDAR-based 3D object tracking and dynamic point removal, is adopted as a baseline

TABLE I
LOCALIZATION ERRORS OF DIFFERENT METHODS ON THE SELECTED KITTI SEQUENCES (UNIT: M)

Sequence	DLO*	TRLO	Ours
05	3.34	2.38	**2.35**
07	1.54	0.93	**0.91**
10	**2.13**	2.77	2.76

The optimal results are indicated by **bold text**, and the suboptimal results are indicated by underlined text.
The * symbol denotes that the result is from [14].



(a) Sequence 05 (b) Sequence 07

(c) Sequence 10

Fig. 3. Trajectory comparison results on representative KITTI sequences.

for comparison. To ensure a fair comparison, none of the methods use external global information or prior maps, and loop closure is also disabled. Unless otherwise specified, all experiments are conducted on a laptop equipped with an Intel i7-11800H CPU, 32 GB RAM, and an NVIDIA RTX 3080 GPU, running Ubuntu 20.04 with the Robot Operating System (ROS) noetic.

2) Evaluation Criteria: The Absolute Trajectory Error (ATE) is employed to evaluate self-localization accuracy, as it reflects the discrepancy between the estimated poses and the ground truth. To assess the overall pose accuracy of the entire trajectory, the root mean square error (RMSE) of the ATE is adopted as the primary evaluation metric, which is computed as follows

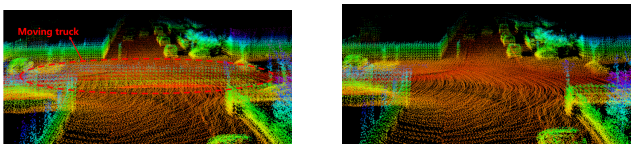
$$e_{RMSE} = \sqrt{\frac{1}{K}\sum_{k=1}^{K} \| \mathrm{x}_k \boxminus \hat{\mathbf{x}}_k^{+} \|_2^2} \tag{9}$$

where x_k and $\hat{\mathbf{x}}_k^{+}$ denote the ground truth and aligned estimated trajectories. K denotes the number of pose measurements in datasets, and $\boxminus$ denotes the subtraction on a manifold.



(a) Sequence 05 (b) Sequence 10

Fig. 4. Illustrations of two-stage visual semantic segmentation–assisted dynamic LiDAR point removal.



(a) Mapping without visual semantic segmentation-assisted dynamic LiDAR point removal

(b) Mapping with visual semantic segmentation-assisted dynamic LiDAR point removal

Fig. 5. Demonstration of mapping consistency. Dynamic LiDAR points are removed by the proposed method, thereby enabling the construction of a consistent and reliable map.

B. Evaluations Under KITTI Dataset

Representative sequences from the KITTI dataset are selected, including typical urban dynamic driving scenarios (Sequence 07), long-range highway scenes (Sequence 05), and mountainous road environments (Sequence 10). As reported in Table I, the proposed method achieves either the best or second-best performance across all sequences, particularly in scenarios with frequent close-range occlusions caused by large dynamic vehicles (Sequences 05 and 07). As illustrated by the trajectories in Fig. 3, the proposed approach also demonstrates a high degree of consistency with the ground truth in local trajectory segments.

Conventional methods such as DLO are unable to effectively handle outlier LiDAR points introduced by dynamic objects, resulting in degraded pose estimation accuracy. Although TRLO is capable of tracking dynamic objects and removing dynamic points to some extent, its detection accuracy for large dynamic objects at close range is limited, leading to incomplete dynamic point removal. As shown in Fig. 4, by leveraging visual semantic segmentation to assist dynamic LiDAR point removal, the proposed method is able to more thoroughly eliminate dynamic LiDAR points, thereby preserving the pose estimation accuracy of the LiDAR odometry.

An ablation study is presented in Fig. 5 to evaluate the effectiveness of the proposed visual semantic segmentation–assisted dynamic LiDAR point removal. Compared with the variant that removes the visual semantic segmentation module, the proposed method is able to accurately identify and eliminate dynamic LiDAR points even in scenarios involving large-scale dynamic objects with close-range occlusions. Without visual assistance, dynamic points caused by such occlusions are often mistakenly retained, leading to ghost artifacts and

inconsistencies in the reconstructed map. In contrast, by leveraging visual semantic cues, the proposed approach achieves more thorough dynamic point removal, resulting in a cleaner and more consistent map. These results demonstrate that visual semantic segmentation plays a critical role in improving mapping consistency in highly dynamic urban environments.

V. Conclusion

This paper presents a LiDAR odometry framework enhanced by visual semantic segmentation for robust self-localization in dynamic urban environments. By combining LiDAR-based 3D object detection with a two-stage visual-assisted dynamic point removal strategy, the proposed method effectively mitigates the adverse impact of moving objects while preserving valid static measurements. The integration of semantic cues enables more thorough elimination of dynamic LiDAR points, especially in challenging scenarios involving close-range occlusions and sparse distant observations. Extensive experiments on real-world datasets demonstrate that the proposed approach achieves superior localization accuracy and robustness compared to existing methods in dynamic scenes. A current limitation of the system lies in the limited camera field of view, which cannot fully overlap with the LiDAR sensing range. Future work will focus on extending the framework to support omnidirectional camera systems, enabling full-surround visual coverage and further improving dynamic point filtering performance.

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